

AllelePerturb enables power-aware evaluation of allele-specific single-cell perturbation prediction

Abstract

Single-cell perturbation models are increasingly used to predict cellular responses to unseen interventions, yet existing benchmarks typically define perturbations at the level of genes, drugs or experimental conditions. This formulation does not test whether models can distinguish protein-coding variants within the same gene, despite the allele-specific nature of many disease mechanisms. Here we present AllelePerturb, a benchmark and evaluation framework for held-out allele-specific response prediction in single-cell transcriptomics. AllelePerturb integrates 470 protein-coding variants across TP53, KRAS, GATA1 and JAK1, comprising 321,043 cells profiled by Perturb-seq, base editing and scSNV-seq. Across feature-based predictors and published variant-conditionable models, transcriptional perturbation direction was recovered, but allele identity was not reliably discriminated. Decomposition of variant responses showed that direction-based metrics predominantly captured gene-shared programs rather than allele-specific residuals. Split-half, sibling-allele and replicate-oracle analyses further showed that benchmark validity depends on whether alleles are reproducibly identifiable above sampling noise, not merely detectable relative to wild type, a discrimination floor present in the single-cell distributions and not only in averaged profiles. TP53 and KRAS were measurement-limited, whereas JAK1 retained a resolvable allele-specific signal but remained computationally unsolved. Controlled predictor analyses linked replicate resolution directly to model-ranking stability, and pilot measurements of 25–50 cells per perturbation predicted future rankability on held-out cells. AllelePerturb therefore provides a power-aware framework for determining when allele-specific perturbation prediction is measurable, benchmarkable and computationally meaningful.

Introduction

Single-cell perturbation prediction has emerged as a central task in virtual-cell modeling, with the goal of forecasting cellular responses to interventions that have not been experimentally observed¹. High-throughput genetic and chemical perturbation screens now provide training data across diverse cell states and intervention classes^{2,3}, motivating generative, optimal-transport, graph-based and foundation-model approaches for response prediction^{4–15}. In parallel, community resources and benchmarks have begun to standardize perturbation analysis and model comparison across datasets, cell types and perturbation settings^{16–18}. These developments have made perturbation prediction an important framework for functional genomics, drug-response modeling and cellular engineering. However, the validity of any benchmark depends on whether its experimental measurements resolve the biological distinctions that models are expected to predict.

Most current perturbation-prediction benchmarks define an intervention at the level of a gene, drug or experimental condition. A model may therefore be asked to predict the response to perturbing TP53, targeting KRAS or applying a compound, but not to distinguish the effects of different protein-coding variants within the same gene. This distinction is biologically and clinically important because disease

mechanisms are often allele-specific. Different TP53 missense variants can disrupt protein folding, DNA binding or regulatory interactions through distinct mechanisms¹⁹, whereas KRAS substitutions at the same or neighboring residues can differ in biochemical activity, downstream signaling and phenotypic effect²⁰. A model that predicts the average response to a TP53 or KRAS perturbation therefore need not distinguish the consequences of individual alleles. This gap is becoming consequential: as single-cell perturbation models are proposed for variant-effect interpretation, therapeutic design and virtual-cell simulation, and as large numbers of variants of uncertain significance await functional characterization, a model that cannot separate alleles of the same gene cannot support these uses. Multiplexed assays of variant effect and deep mutational scanning have enabled systematic evaluation of protein variants^{21,22}, but these assays generally summarize fitness, activity or molecular function as scalar outcomes rather than population-level single-cell transcriptional responses.

Variant-resolved single-cell assays and variant-conditioned models have begun to make allele-specific response prediction experimentally accessible. Single-cell phenotyping has shown that coding variants in TP53 and KRAS can produce distinct transcriptional responses when measured at sufficient depth²³, and PerturbNet introduced continuous variant representations into conditional response prediction for GATA1 variants²⁴. At the same time, recent studies have shown that perturbation-prediction results can be strongly influenced by simple baselines, systematic variation and the geometry of the evaluation metric^{1,16–18,25,26}. These observations expose a more fundamental problem for allele-level evaluation. Standard metrics such as mean squared error or Pearson correlation can reward recovery of a gene-shared transcriptional program without establishing that a model distinguishes individual alleles. A predicted response may point in the correct overall direction while remaining equally compatible with several variants of the same gene.

Allele-level evaluation therefore requires three nested forms of resolution. First, a variant must be detectable relative to wild type. Second, it must be reproducibly identifiable from other variants of the same gene. Only then is it meaningful to ask whether the response to a held-out allele can be predicted from its molecular features. Detection is necessary but not sufficient: multiple alleles may all differ from wild type while remaining indistinguishable from one another. When within-variant sampling variability approaches the transcriptional separation between alleles, the measured ground truth contains little information with which to reward an allele-specific prediction. Under this regime, apparent model failure may reflect insufficient measurement resolution rather than inadequate modeling, and a benchmark may fail to recover the correct ordering of models even when their underlying predictive quality differs.

Here we introduce AllelePerturb, a benchmark and evaluation framework for held-out protein-coding variant prediction in single-cell transcriptomics. AllelePerturb-Bench integrates 470 variants across TP53, KRAS, GATA1 and JAK1, comprising 321,043 cells profiled by Perturb-seq, base editing and scSNV-seq^{23,24,27}. AllelePerturb-Eval separates prediction quality into transcriptional direction recovery, allele discrimination and differential-expression fidelity, and evaluates generalization across random, positional, mechanistic, low-depth and protocol-compatible splits. Across feature-based predictors and published models that can condition on continuous variant representations, perturbation direction was recovered but allele identity was not reliably discriminated. Decomposition of the response profiles showed that direction-based performance was dominated by gene-shared programs rather than allele-specific residuals. Split-half, sibling-allele and replicate-oracle analyses further distinguished measurement-limited regimes, in which allele-specific performance could not be reliably evaluated, from model-limited regimes, in which the measurement was adequate but current predictors remained unsuccessful. Controlled predictors linked replicate resolution directly to the stability of model rankings, and pilot measurements of 25–50 cells per perturbation predicted future rankability on held-out cells. Together, these analyses establish a power-aware framework for determining when allele-specific perturbation prediction is measurable, benchmarkable and computationally meaningful. More broadly, AllelePerturb introduces a measurement-power-aware evaluation principle, testing whether the experimental ground truth can reward a model before models are ranked against it, that applies to perturbation benchmarks well beyond coding variants.

Results

The AllelePerturb pipeline for held-out allele prediction

AllelePerturb formulates held-out allele prediction as an end-to-end pipeline (Fig. 1a). Each perturbation is a protein-coding variant measured at single-cell resolution, and its population-average response, the pseudobulk difference δ_v between the variant's cells and wild-type cells, is the prediction target. A model receives only the variant's molecular features, a six-dimensional biophysical vector θ or a protein-language-model embedding, and never the variant identity or its cells; it predicts $\hat{\delta}_v$ for an unseen allele. Predictions are scored on two axes: direction recovery (Pearson- δ , whether $\hat{\delta}_v$ points the right way in expression space) and allele discrimination (PDS, whether $\hat{\delta}_v$ is closest to its own variant among the sibling variants of the gene). In parallel, and directly on the single-cell distributions rather than on the mean, a resolution diagnostic asks whether a variant is separable from wild type (detection) and from its siblings (identification), that is, whether its ground truth can reward an allele-specific model at all. The pipeline therefore returns not only a model score but a power-aware verdict on whether a perturbation is measurable, benchmarkable and worth modeling. This design targets a distinction that gene-level prediction cannot make: in TP53, a gene-level representation collapses variant responses into a common direction, whereas allele resolution preserves variant-specific departures from wild type, with mutations such as R175H, R273C and R248Q occupying distinct regions of the response space.

AllelePerturb-Bench integrates four variant-perturbation datasets spanning different proteins, cell types and experimental technologies (Fig. 1b,c). TP53 (393 aa, 98 variants) and KRAS (189 aa, 92 variants) were profiled by Perturb-seq in A549 cells²³, GATA1 (413 aa, 254 variants) by base editing in hematopoietic stem and progenitor cells²⁴, and JAK1 (1,154 aa, 26 variants) by scSNV-seq in HT-29 cells²⁷. Together, these datasets comprise 470 protein-coding variants and 321,043 single cells, including the dataset wild-type and control cells. Per-variant cell depth varies substantially across genes, with median depths of 929 and 1,000 cells per variant for TP53 and KRAS, 355 for GATA1 and 104 for JAK1 (Fig. 1c), providing a natural range of sampling regimes for testing variant-level evaluation.

To make held-out variant prediction possible, each variant was represented by protein-level features rather than by a variant identifier. We encoded variants using a six-dimensional biophysical vector θ , including changes in hydrophobicity, side-chain volume and charge, together with fold-core location, functional-switch residue status and hotspot/pathogenic annotation (Fig. 1d). These features describe variant properties beyond gene identity and allow variants from different genes to be placed in a shared feature space. AllelePerturb-Eval then separates prediction quality into three dimensions: direction recovery, allele-level discrimination and differential-expression fidelity. Pearson- δ measures agreement between predicted and observed perturbation directions, whereas PDS directly tests the defining allele-level requirement: whether a predicted response identifies its target variant among other held-out variants (Fig. 1e). Six split strategies further test random generalization, positional extrapolation, mechanistic extrapolation, cross-gene transfer, low-depth evaluation and compatibility with existing protocols (Fig. 1f). This separation of direction recovery from allele discrimination sets up the central question of the benchmark: whether models can distinguish alleles, rather than merely recover the shared direction of a perturbed gene.

Variant-level prediction exposes a direction-ranking dissociation

Having defined allele-level discrimination as the central requirement of the task, we next asked whether predictors that recover the direction of a perturbation also identify the correct held-out variant. We evaluated 20 predictors or references, comprising 18 feature-model combinations (six model heads by three feature spaces), the Gene-mean predictor and the WT-null reference (Fig. 2). Across the 19 non-null predictors, allele discrimination remained at chance: PDS-cosine clustered between 0.49 and 0.52 (mean 0.50), and no predictor's 95% bootstrap confidence interval excluded the 0.50 chance level (Fig. 2b). Point estimates were averaged over 15 independent pseudobulk subsamples to remove single-draw variability, and the interval of every predictor overlapped chance. This conclusion did not depend on the analytic 0.50 baseline: within-split label permutation gave an empirical null mean of 0.500 (range



Figure 1: The AllelePerturb pipeline for held-out allele prediction. **a**, The AllelePerturb pipeline. A held-out protein-coding variant is measured at single-cell resolution; its pseudobulk response δ_v (variant cells minus wild-type cells) is the prediction target. A model maps the variant's molecular features (θ or a protein-language-model embedding), not its identity or cells, to a predicted profile $\hat{\delta}_v$, scored for direction recovery (Pearson- δ) and allele discrimination (PDS, retrieval of the variant among its siblings). In parallel, a single-cell resolution diagnostic (detection versus wild type and identification versus siblings, computed on the cell distributions) determines whether the ground truth can reward an allele model, yielding a power-aware verdict on whether a perturbation is measurable, benchmarkable and worth modeling. **b**, Benchmark coverage: variant positions along the four proteins (TP53, KRAS, GATA1 and JAK1) with annotated domains and hotspot residues; AllelePerturb-Bench integrates 470 protein-coding variants across three experimental technologies (Perturb-seq, base editing and scSNV-seq). **c**, Per-variant cell depth varies substantially across genes, providing a range of sampling regimes for evaluation (98, 92, 254 and 26 variants; 83.4k, 83.6k, 149.2k and 4.9k cells, including wild-type/control, for TP53, KRAS, GATA1 and JAK1, respectively). **d**, Each variant is represented by a six-dimensional biophysical vector θ encoding changes in hydrophobicity, side-chain volume and charge, together with fold-core location, functional-switch residue status and hotspot/pathogenic annotation, placing variants from all genes in a shared feature space. **e**, AllelePerturb-Eval separates prediction quality into direction recovery (Pearson- δ), allele-level discrimination (PDS) and differential-expression fidelity. **f**, Six generalization splits test random, positional-extrapolation, mechanistic-extrapolation, cross-gene, low-depth and compatibility

0.495 to 0.504 across 340 method-split-gene combinations), and only 3.2% of combinations exceeded the permutation 95th percentile, consistent with chance under the empirical permutation null. PDS-L1 and PDS-L2 gave the same conclusion.

In contrast, the same predictors consistently recovered the transcriptional direction of perturbation effects. Pearson- δ ranged from 0.55 to 0.65 across predictors, and every non-null predictor had a 95% confidence interval entirely above zero (Fig. 2c). Plotting PDS against Pearson- δ placed all predictors in the same regime: direction recovered, allele ranking failed (Fig. 2d). Method-to-method variation was small compared with the separation between these two evaluation axes, indicating that variant-level prediction exposes a systematic direction-ranking dissociation rather than a failure of a particular model class.

The recovered direction was almost entirely gene-shared. When each variant's effect was decomposed into a gene-shared component and an allele-specific residual, $\delta_v = \bar{\delta}_g + r_v$, the residual Pearson- δ was near zero for every predictor (between -0.08 and 0.03), and a featureless predictor that outputs only the training gene mean already reached the same direction score as the best feature model. Removing the shared component removed the recovered signal, so the models capture the program-level response of a gene rather than the allele-specific residual that distinguishes its variants. Consistent with this, the allele-specific residual lay near the floor for the low-effect genes: an oracle scored on residual profiles remained at chance for TP53 and KRAS (0.48) and only marginal for GATA1 (0.59), rising clearly only for JAK1 (0.89), whose variants carry a resolvable allele-specific signal (Supplementary Note).

Differential-expression fidelity provided a biological interpretation of the same dissociation (Fig. 2e). Direction agreement reached 65%, whereas DE-LFC Spearman was 0.25 and DE overlap was only 15%. This gradient (direction 65%, LFC ranking 25%, gene-set overlap 15%) suggests that predictors learn a program-level perturbation direction but progressively lose the finer signal needed to rank effect sizes and identify variant-specific response genes.

The magnitude of the dissociation, defined per gene as the mean Pearson- δ minus the mean PDS pooled over splits, varied by gene (Fig. 2f). It was strongest for TP53 (0.31), intermediate for KRAS (0.19) and GATA1 (0.13), and near zero for JAK1 (-0.04). This ordering anticipated the split-half signal-to-noise analysis below: JAK1 had the strongest per-variant signal and the weakest dissociation (its two axes nearly coincide because its per-gene direction recovery is itself lower, consistent with a less gene-shared, more allele-individuated response), whereas TP53 and KRAS lay near the noise floor. Because each gene here is also a distinct dataset, assay and cell type, this ordering is a per-dataset gradient rather than an intrinsic property of the genes (see Discussion). At the single-variant scale, the same pattern appeared within TP53: Ridge-esm produced stable Pearson- δ values across representative held-out variants (0.74 to 0.82), but PDS varied from 0.00 to 0.98, with roughly half of examples below chance (Fig. 2g). These results suggest that allele-ranking failure is not explained by model choice alone and motivate a direct test of whether the ground truth itself is measurable at variant resolution.

Allele identification, not detection, sets the ground-truth resolution floor

The direction-ranking dissociation raised a mechanistic question: do predictors fail to rank alleles because the models are wrong, or because the ground-truth variant responses are not stably separable at the available depth? Held-out allele prediction requires three nested resolutions (Supplementary Table 3): *detection* (a variant is distinguishable from wild type), *identification* (a variant is distinguishable from its sibling variants) and *prediction* (a held-out variant is recoverable from its features). Detection is necessary but not sufficient; the resolution that governs whether a benchmark can score allele models is identification, because a metric can only reward a model for separations that the ground truth itself resolves. We measured the detection window with a split-half reproducibility analysis: for each variant, cells were randomly divided into two halves, and we compared the energy distance between the two halves of the same variant (D_{self}) with the distance between that variant and wild type (D_{null} ; Fig. 3a,b). A $D_{\text{self}}/D_{\text{null}}$ ratio near 1 indicates that within-variant sampling variation is comparable to the variant-to-wild-type signal; a ratio well below 1 indicates a resolvable detection window.

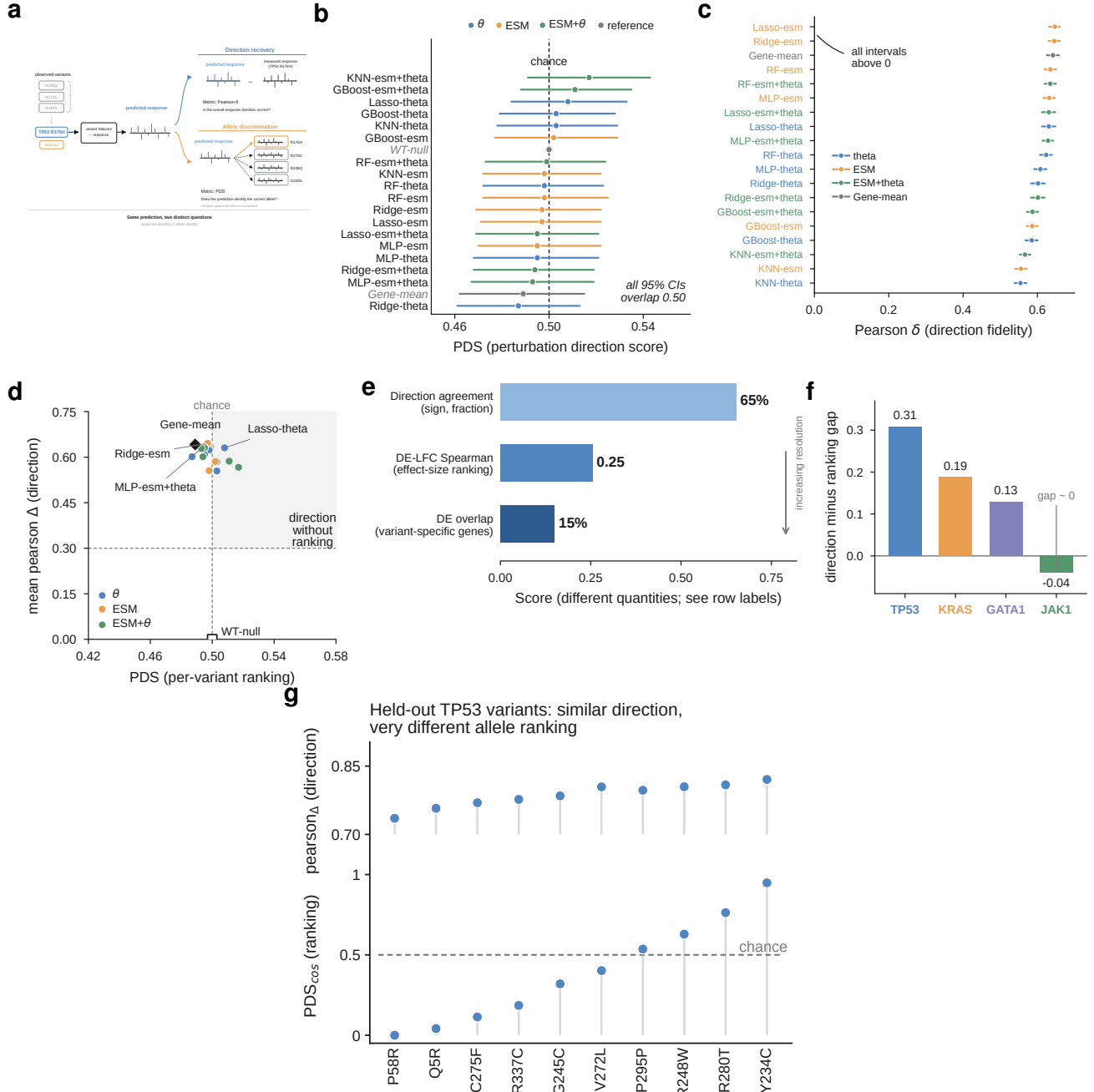


Figure 2: Feature-based predictors recover gene-shared direction without reliable allele discrimination. **a**, Unified evaluation of a held-out variant: the predicted response is scored for direction recovery (Pearson- δ) and for allele discrimination (PDS, whether it identifies its own variant among the other held-out variants). **b**, Allele-level PDS-cosine for 20 predictors with 95% bootstrap confidence whiskers; every interval overlaps the chance level of 0.50. **c**, The same predictors recover perturbation direction (Pearson- δ), with every interval above zero. **d**, Methods cluster in the direction-without-ranking regime. **e**, Differential-expression fidelity: direction agreement 65%, DE-LFC Spearman 0.25, DE overlap 15%. **f**, The dissociation gap (per-gene mean Pearson- δ minus mean PDS-cosine, pooled over splits) follows a per-dataset gradient across the four genes (TP53 0.31, KRAS 0.19, GATA1 0.13, JAK1 -0.04). **g**, Per-variant scores (Ridge-esm, TP53) show stable direction but unstable ranking.

By this detection measure the four genes occupied distinct regimes (Fig. 3c): TP53 ($D_{\text{self}}/D_{\text{null}} = 0.96$, 95% CI 0.95–0.98), KRAS (1.00, 0.99–1.02) and GATA1 (0.88, 0.85–0.90) sat near the floor, whereas JAK1 was well separated from it (0.21, 0.12–0.33), the same gene ordering seen in the direction-ranking dissociation of Fig. 2f. Detection was at best marginal for the three low-effect genes: their variant-to-wild-type shift is comparable to within-variant replicate noise ($D_{\text{self}}/D_{\text{null}} \approx 1$), so although a weak split-half criterion counts a slim majority as detectable (Fig. 3g), the signal barely clears the per-variant floor. Identification, the stricter resolution that actually bounds allele benchmarking, then failed. A model-free classifier trained to separate the single cells of two sibling variants under cross-validation was at chance for TP53 and KRAS and only marginal for GATA1 (median identification AUROC 0.49, 0.50 and 0.52, against 0.50 under label permutation and a wild-type-versus-wild-type control), rising only for JAK1 (0.87; Fig. 3e). Distributional distances agreed: the transcriptional distance between sibling alleles fell below within-variant replicate noise for almost every pair in TP53, KRAS and GATA1 (median nearest-sibling cosine distance 0.23 to 0.46 versus replicate noise near 0.97), leaving 0%, 0% and 3.8% of allele pairs resolvable, whereas the threefold-lower replicate noise of JAK1 left 78.5% resolvable (Fig. 3d; at the single-nearest-sibling level JAK1 identification is a benchmark-level rather than a per-variant property). Detection and identification therefore degraded together across these datasets, but they are distinct rungs of the same ladder: detection-level rankability, the split-half signal-above-replicate-noise criterion used for Fig. 3g,h and for the atlas comparison below, is necessary but not sufficient for identification, which is the resolution that actually bounds allele benchmarking because a benchmark can only reward a model for separations that the ground truth itself resolves. Only JAK1 cleared identification. This identification verdict was invariant across five distance families, from energy and maximum-mean-discrepancy to sliced-Wasserstein, cosine and Euclidean, so it is not an artifact of energy-distance geometry (Supplementary Note). The quantity that bounds allele benchmarking is therefore not raw cell depth but the effect-size-to-noise window available for identification. Because the classifier and these distributional distances operate directly on the single cells, they constitute a single-cell control for the pseudobulk prediction target: the discrimination floor is present in the full cell distributions, not only in averaged profiles, so the mean target is not the limiting factor (Supplementary Note).

This point was clearest when effect size and depth were considered together (Fig. 3f,g). JAK1 had the lowest nominal depth among the four genes (median 104 cells per variant) but the widest effect-size-to-noise window, and it was already saturated in the split-half detection assay at 50 cells per half. In contrast, TP53 and KRAS had much higher nominal depths but narrow allele-specific effect windows, remaining close to the noise floor even with median raw depths of 929 and 1,000 cells per variant. Sub-sampling over 50 to 300 cells per half confirmed the same ordering: JAK1 remained fully detectable across the range, whereas TP53, KRAS and GATA1 increased only modestly and plateaued below 75%. Within the tested range of 50 to 300 cells per half, additional cells improved detection only modestly for TP53, KRAS and GATA1 and left JAK1 already saturated, indicating that depth alone is insufficient when the allele-specific effect is small relative to sampling and biological variability, rather than establishing a universal depth law.

The detection floor was not specific to allele-level data. Under the same native-depth, all-perturbation, direct split-half energy-distance definition, the three low-effect allele-level genes (TP53, KRAS and GATA1) were 98–100% un-rankable by this detection-level split-half criterion, whereas JAK1 remained a positive control with only 10% un-rankable variants (Fig. 3h). Among gene-level atlases, Replogle 2022² showed 55.3% un-rankable perturbations (95% CI 52.8–57.5%), compared with 14.6% for Adamson 2016²⁸ and 3.4% for Norman 2019²⁹ (Fig. 6f). Results were qualitatively robust under alternative rankability criteria, including bootstrap-CI-based, increased-stringency and ratio-threshold definitions (Supplementary Fig. 1). Thus, the detection floor is not a peculiarity of allele-level data: finite-depth distributional distances can leave a substantial fraction of individual perturbations below the resolution required to score them. This per-perturbation floor is distinct from the benchmark-level ranking resolution examined next, to which many individually un-rankable perturbations can still contribute in aggregate (Fig. 5).

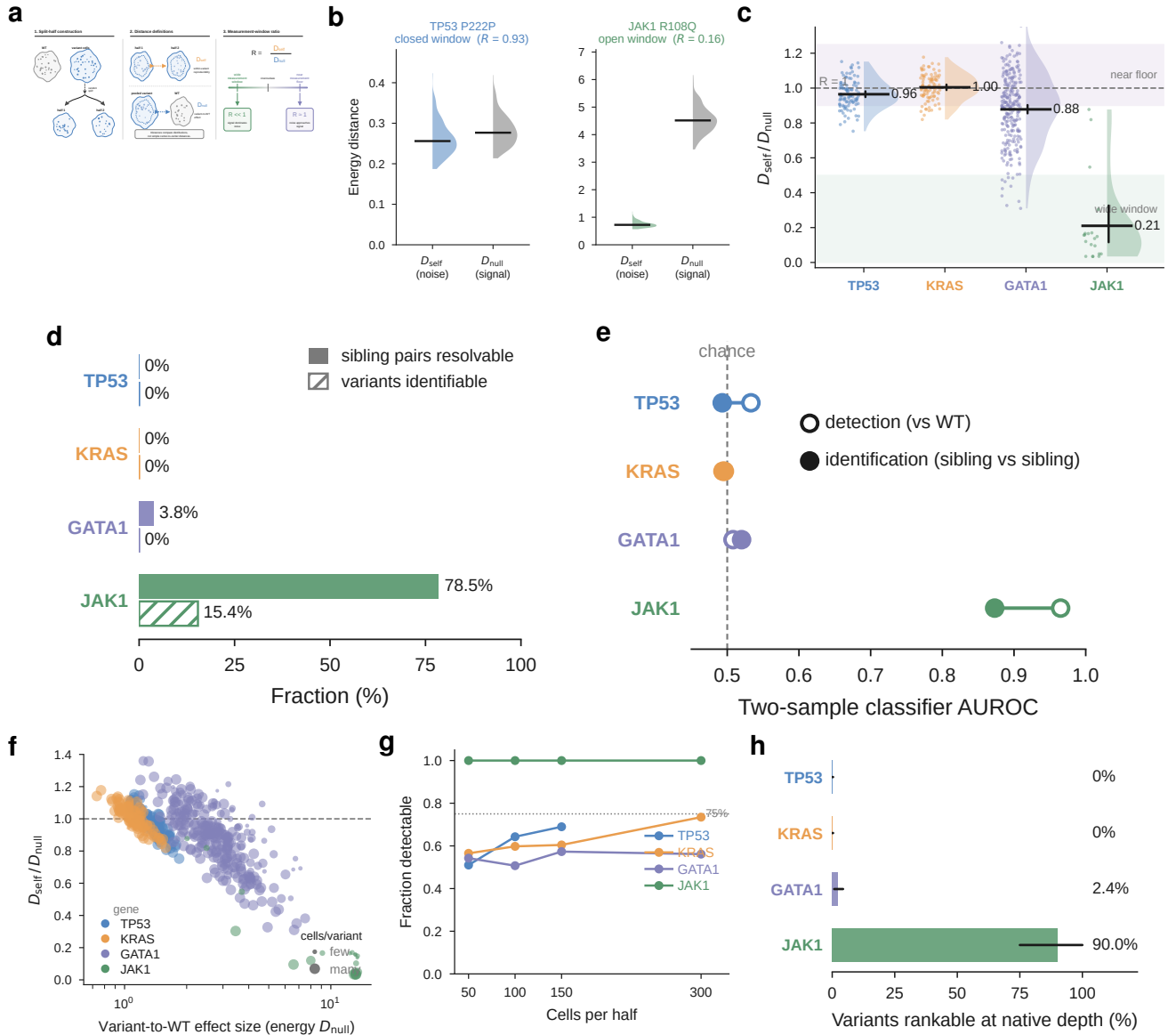


Figure 3: Split-half analysis reveals a marginal detection window and absent allele identification. **a**, Definition of the detection window from within-variant (D_{self}) and variant-to-wild-type (D_{null}) distances. **b**, Representative variants illustrate a closed window (TP53 P222P, $D_{self} \approx D_{null}$) and an open window (JAK1 R108Q, $D_{self} \ll D_{null}$). **c**, Per-variant D_{self}/D_{null} places TP53 (0.96), KRAS (1.00) and GATA1 (0.88) near the detection floor and JAK1 (0.21) well below it. **d**, Allele identification: the fraction of sibling-variant pairs whose transcriptional distance exceeds within-variant replicate noise (TP53 and KRAS 0%, GATA1 3.8%, JAK1 78.5%) and the stricter fraction of variants whose single nearest sibling is separable (JAK1 15.4%, 0% for the other genes); detection is necessary but not sufficient for identification, the resolution that bounds allele benchmarking. **e**, Classifier two-sample test: cross-validated AUROC separating the single cells of sibling variants (identification) is at chance for TP53 and KRAS (0.49, 0.50), marginal for GATA1 (0.52) and high for JAK1 (0.87); detection against wild type (open symbols) is likewise near chance for the low-effect genes and high only for JAK1; label permutation gives 0.50. **f**, Effect size and effective depth jointly determine the window, not cell number alone. **g**, Depth titration: detection rate rises with cells per half in a gene-dependent manner (JAK1 saturated at 50 cells, others plateau below 75%). **h**, Fraction of variants detection-rankable at native depth, i.e. split-half signal above replicate noise (TP53 0%, KRAS 0%, GATA1 2.4%, JAK1 90.0%); gene-level atlases are compared in Fig. 6f.

Allele-ranking failure persists across evaluation choices and model interfaces

We next tested whether the allele-ranking bottleneck could be explained by evaluation design or variant representation rather than by the measurement regime identified above. Across five of the six generalization splits (the sixth, cross-gene, is defined but excluded from the main grid, Methods, because the datasets do not share a gene space), Pearson- δ and PDS behaved differently (Fig. 4a,b). Pearson- δ remained positive under random, positional-extrapolation and compatibility splits, typically between 0.55 and 0.75, and declined mainly under mechanistic extrapolation and low-depth evaluation. By contrast, PDS remained near the empirical chance regime across all splits (Fig. 4a,c), with the strongest degradation under the low-depth split. No split produced a consistent above-chance allele-discrimination regime, indicating that the direction-ranking dissociation was not specific to the random holdout setting.

The conclusion was also insensitive to the distance function used for PDS (Fig. 4d). Cosine, L1 and L2 distances yielded nearly identical PDS distributions across the 18 feature-model combinations, all clustered near the chance level of 0.50. Shifting from methods to gene-by-split combinations gave the same picture (Fig. 4f): Pearson- δ exceeded PDS in 15 of 17 combinations, with the two exceptions occurring in small held-out mechanistic-extrapolation sets where both metrics were close. The GATA1 low-depth split had a PDS of 0.19, consistent with the detection-limit prediction from the split-half analysis. The conclusion was equally robust to preprocessing: refitting the per-gene standardization on training variants and wild-type cells only, excluding every held-out variant, left all predictors at or below chance (no method above 0.50; mean Δ PDS = -0.01 against the reported all-cell standardization) with unchanged direction recovery, so the transductive standardization used throughout is, if anything, conservative with respect to the null (Supplementary Note).

Richer variant representations did not close the gap either. Under the identical splits, heads and scoring, we compared the six-dimensional biophysical θ vector, 1,280-dimensional ESM-1v embeddings, modern ESM2 embeddings³⁰, three mutation-aware ESM2 constructions that encode the change relative to wild type (the mutant-minus-wild-type global embedding, the mutated-residue hidden-state difference and a local-window difference), and AlphaMissense³¹, a purpose-built structure- and evolution-informed variant-effect predictor. Every representation left allele discrimination at chance (PDS 0.46 to 0.48, single-draw estimates that sit marginally below the 15-subsample values of Fig. 2 but carry the same chance verdict) with similar direction recovery (Pearson- δ 0.63 to 0.66; Fig. 4e, Supplementary Note). The direction channel was not feature-driven: a featureless predictor that outputs the training gene mean already reached Pearson- δ 0.66, and shuffling the ESM2 features across variants within a gene left PDS at chance, confirming that the recovered direction is the shared gene-level program rather than allele-specific signal. The failure was also not a matter of the representation discarding the mutation: the mutated-residue difference was highly variant-specific, with within-gene distances 91% of between-gene distances, against 0.1% for the gene-identity-dominated global embedding, yet it too failed. AlphaMissense pathogenicity did not track the measured transcriptional effect size within any gene (Spearman near zero for TP53, KRAS and GATA1). Thus more modern, mutation-aware and structure-aware representations recover the shared perturbation direction but do not overcome the allele-discrimination floor. For the low-effect genes this floor lies in the ground-truth measurement rather than in the input; for JAK1 the input is likewise not the bottleneck, and the residual gap is one of computational generalization, in that the evaluated feature-to-response mappings do not recover an allele signal that the measurement resolves (Fig. 5).

Published perturbation-prediction models fared no better than the feature-regression heads under a matched evaluation. We re-scored external models through a single harness: the same held-out variants, pseudobulk profiles, tie-aware PDS and bootstrap and permutation controls used for the in-house heads (Fig. 4g, Supplementary Table 4; Methods). Four of the five variant-conditionable models ran to completion, and all remained at chance: scGen (PDS 0.49, 95% CI 0.46–0.52), scVIDR (0.50, 0.48–0.53), Biolord (0.49, 0.47–0.52) and CellFlow (0.51, 0.48–0.54). None exceeded the feature-model baselines of Fig. 2, whose intervals also overlapped chance. The remaining public tools cannot represent the task: scGPT,

GEARS and STATE condition on perturbed-gene identity, so all variants of a gene collapse to a single prediction and only 4 of 470 variants (0.85%) are representable (Fig. 4h). Consistent with this, the gene-constant Gene-mean reference sits at chance (PDS 0.49). A compositional autoencoder (CPA) fitted with a variant-feature encoder did not converge on the continuous inputs. One generative flow, PerturbNet, marginally exceeded the analytic 0.50 chance level (PDS 0.54), but its predictions are confined to a 50-dimensional expression subspace in which random vectors already score 0.52; against a permutation null estimated in that subspace its exceedance was not significant (permutation $P = 0.13$), and its direction recovery was negligible (Pearson- δ 0.13). Five model classes that can represent an allele were scored on the task, feature regression, latent arithmetic, disentanglement, optimal transport and conditional-flow generative models, and none achieved above-chance allele discrimination; the gene-keyed foundation models could not be scored at all (an interface incompatibility, not a performance failure) and the compositional autoencoder did not converge. The consistency of this chance-level result across every model class that can represent the task argues against a failure specific to any single architecture and motivates a direct test of whether the ground truth itself resolves alleles at the available depth (Fig. 5).

Replicate resolution governs attainable discrimination and benchmark validity

A benchmark can reward a model only for signal that the measurement itself resolves. To bound what is achievable, we computed an oracle predictor whose prediction for each held-out variant is an independent second measurement of that same variant, taken from a disjoint split-half pseudobulk and scored through the identical harness; this defines an empirical replicate-based discrimination ceiling at the specified depth and protocol (Methods). For TP53 and KRAS the ceiling was itself at chance (PDS 0.485, 95% CI 0.445–0.525, and 0.500, 0.440–0.566). This was not an artifact of split-half depth: it was unchanged when all cells were used (0.486 and 0.492), and even a perfect prediction, the deep true mean of a variant, scored against the benchmark’s own noisy measurement remained at chance (0.49 and 0.49; Supplementary Note). Above-chance allele discrimination therefore could not be reliably demonstrated at this resolution: the oracle ceiling, the best feature-regression head (a maximum over 18 models, upward-biased under the null) and a perfect-prediction control are all statistically indistinguishable from 0.50 with overlapping 95% confidence intervals. Because even a perfect prediction, scored against the benchmark’s own noisy measurement, stayed at chance, the binding limit lies in the resolution of the ground truth rather than in model capacity. This replicate ceiling is an empirical reproducibility reference at the available depth and protocol, not a mathematical upper bound on all denoising or hierarchical models; a model that pooled information across variants could in principle exceed a single replicate, but the benchmark as measured could not reward it. GATA1 sat marginally above chance (0.572, 0.549–0.595), matching its small resolvable-pair fraction. For JAK1, by contrast, the ceiling reached 0.792 (0.742–0.838) while the best in-house model stayed near chance (0.52), revealing a genuine computational gap in which the measurement is adequate but current models do not exploit it (Fig. 5b,c, Supplementary Table 5). Consistent with the confounds noted above, the higher JAK1 ceiling reflects its more favorable measurement regime rather than necessarily stronger allele biology.

The same replicate resolution governs a distinct, benchmark-level quantity: whether the benchmark can order models at all. We built synthetic predictors of known, graded quality by interpolating between the gene-mean profile and each variant’s measured effect (Fig. 5d,e), and asked how often the observed PDS recovered their true ranking under bootstrap resampling of the held-out variants (Methods). When the oracle window was at chance (ceiling ≤ 0.52 ; TP53 and most depths of KRAS), the benchmark recovered the true predictor order only 27% of the time and identified the best predictor only 38% of the time, far from reliable. Once the window opened even modestly (ceiling ~ 0.57 ; GATA1), correct ordering rose to 95% and correct winner selection to 97%; above a ceiling of 0.65 both approached 1.0 (Fig. 5g); these transition points are empirical to the present candidate-set sizes and distance metric, not universal thresholds. Recovery also fell when few variants were held out, matching the expected dependence on test-set size. Rankability is therefore not merely descriptive: below this resolution a benchmark cannot reliably distinguish a good model from a bad one, and any leaderboard it produces is unstable regardless

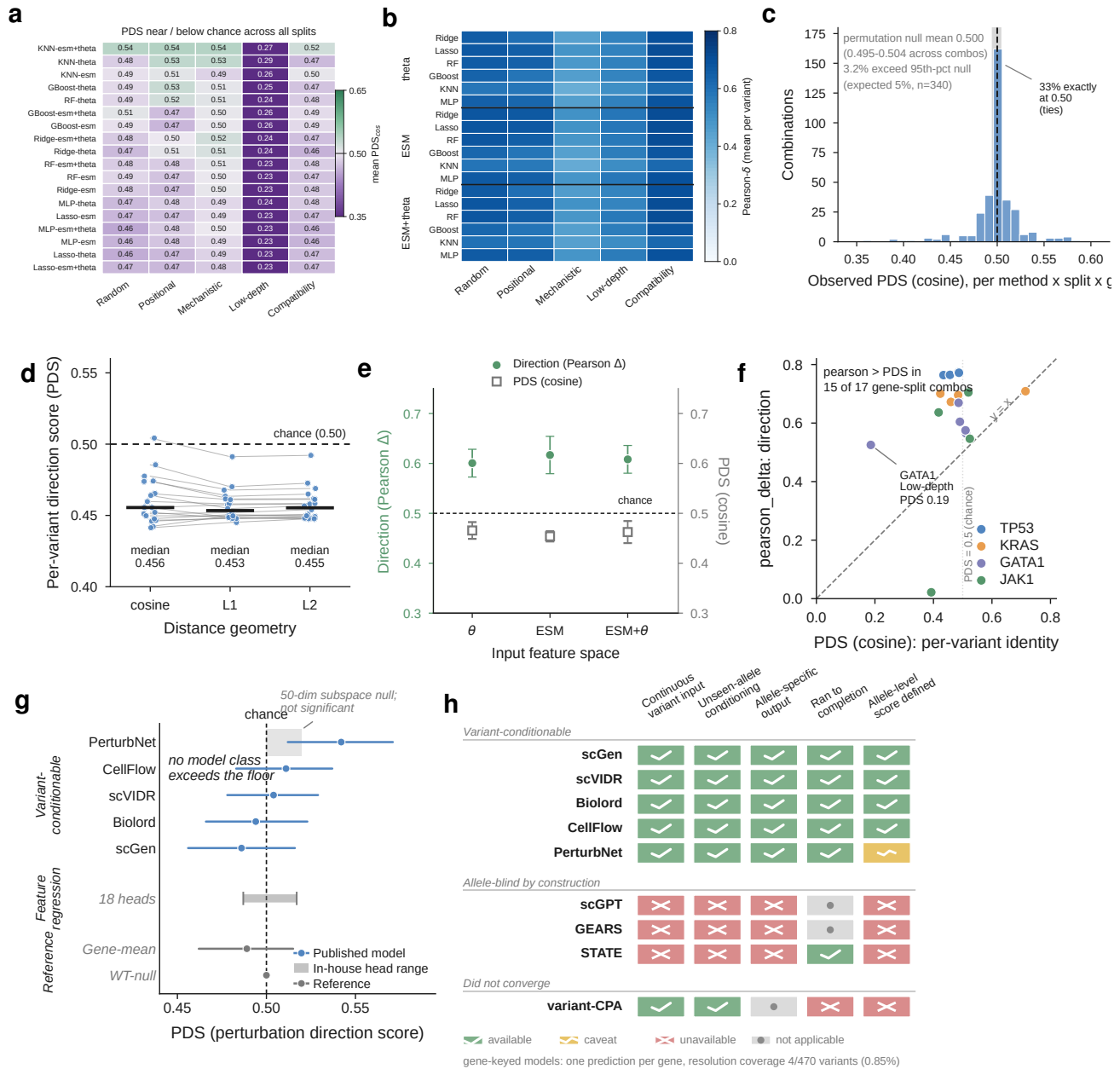


Figure 4: Allele-ranking failure persists across evaluation choices and model interfaces. **a**, PDS across five generalization splits stays near chance, with no split showing a consistent above-chance allele-discrimination regime and the strongest collapse under the low-depth split. **b**, Pearson- δ stays consistently positive across the same splits. **c**, Empirical permutation null: within-split label permutation centers PDS at 0.50 (per-combination null means 0.495–0.504), with observed exceedances of the 95th percentile at 3.2% (consistent with chance), so chance-level PDS is not an artifact of the analytic baseline. **d**, Ranking failure is invariant to the distance function (cosine, L1 and L2 all near chance across the 18 feature-model combinations). **e**, Richer features do not close the ranking gap (θ , ESM and ESM+ θ all near 0.50). **f**, The dissociation holds across all gene-by-split combinations, with Pearson- δ exceeding PDS in 15 of 17. **g**, Published perturbation models scored under one identical harness (same held-out variants, pseudobulk profiles, tie-aware PDS and bootstrap/permutation controls): every variant-conditionable model except PerturbNet, the in-house feature-regression range and both references overlap the PDS = 0.50 chance line; PerturbNet's point estimate (0.54) exceeds chance in the full harness but falls inside its own 50-dimensional subspace null (0.52) (Supplementary Table 4). **h**, Interface-compatibility audit: gene-keyed models (scGPT, GEARS, STATE) are allele-blind by construction, resolving only 4 of 470 variants (0.85%); this is interface incompatibility, distinct from performance failure.

of the models' true quality (Fig. 5g). This closes the loop from the split-half diagnostic to benchmark validity and motivates predicting rankability before a screen is run. Applying the same resolution analysis to five diverse public perturbation benchmarks, at a fixed shallow depth of 50 cells per half for comparability, placed this floor in context: Norman, Replogle, Adamson and the Virtual Cell Challenge training set all recovered the true predictor ordering (resolution 1.00) and a drug screen (sci-Plex³²) reached 0.77, whereas TP53 and KRAS remained the only datasets at the floor (0.01 and 0.02; Fig. 5f). This benchmark-level ranking resolution and the per-perturbation un-rankable fraction of Fig. 6f are distinct quantities: an atlas such as Replogle can leave roughly half of its individual perturbations un-rankable yet still order graded predictors perfectly, because resolution pools information across thousands of perturbations, whereas each allele gene contributes at most a few hundred. Their un-rankability therefore reflects not a peculiarity of AllelePerturb but a genuinely lower ground-truth resolution than that of diverse gene- and drug-level screens; consistent with the dependence on test-set size, the many variants of GATA1 (0.51 at this shallow depth, versus 0.95 at its native ceiling in Fig. 5g) partly offset its low per-variant signal.

Rankability prediction enables power-aware evaluation of variant perturbation screens

If allele discrimination is meaningful only above a measurement floor, then variant-level benchmarks need a way to identify which perturbations are evaluable before full model comparison. We therefore asked whether rankability could be predicted from pre-modeling, pilot-estimable quantities. Using leave-one-dataset-out (LODO) cross-validation across the four AllelePerturb datasets and three external Perturb-seq atlases, we trained a rankability predictor that uses each perturbation's pilot-estimable pseudobulk effect size as its single feature (Fig. 6a; Methods).

Across the five datasets in which rankability is non-degenerate, effect size separated rankable from un-rankable perturbations with a mean AUROC of 0.96 (Replogle 0.97, Norman 0.93, Adamson 0.93, GATA1 0.95, JAK1 1.00; ROC curves and 95% bootstrap intervals in Fig. 6b,c). TP53 and KRAS could not be evaluated: at native depth every one of their variants is un-rankable, a single-class outcome that is itself the strongest statement that these low-effect genes lie entirely below the measurement floor. Effect size is therefore a usable triage signal exactly where rankability varies; where it does not, the floor has already foreclosed the task. Because this predictor's feature and label derive from the same cells, we also validated it prospectively on disjoint data: from a pilot of only 25 to 50 cells per perturbation we predicted rankability at higher depth on held-out cells, under leave-one-dataset-out cross-validation. Across the three benchmarks with balanced labels (Adamson, Norman and Replogle) the pilot predictor reached AUROC 0.85, 0.94 and 0.99 (mean 0.93), with near-diagonal calibration and a label-permutation null of 0.49; a training-free predictor using the pilot signal-to-noise ratio performed comparably (0.84 to 0.98 per dataset) and the result held at a 25-cell pilot (Fig. 6d). Rankability is therefore prospectively predictable from a small pilot, before a full screen is committed. This pilot triage estimates perturbation-versus-reference (detection-level) rankability, which is necessary but not sufficient for allele benchmarking: a perturbation can clear the detection floor yet still lack the sibling-level separation that identification requires (Fig. 3d,e), so pilot rankability excludes clearly unmeasurable perturbations rather than guaranteeing allele-level evaluability.

The same analysis also reframes experimental depth as an effect-size-conditioned design variable (Fig. 6e). A strong-effect gene such as JAK1 crossed the detection floor from as few as 50 cells per half, whereas TP53 and KRAS remained near the noise floor despite median raw depths of 929 and 1,000 cells per variant. The relevant design target is therefore not a fixed cells-per-variant threshold, but the depth required for a given effect-size, inter-variant-difference and sampling-noise structure. In this sense, the bottleneck shifts from raw cell count to the effect-size-to-noise window that the experiment must clear.

As a practical check, we revisited the external gene-level atlases in an experimental-design context (Fig. 6f). At native depth, 55% of perturbations were un-rankable in Replogle, compared with 15% in Adamson and 3% in Norman. When normalized to the same shallow depth ($n = 50$ per half), the un-rankable fractions shifted toward one another: Replogle remained high at 55%, while Adamson in-

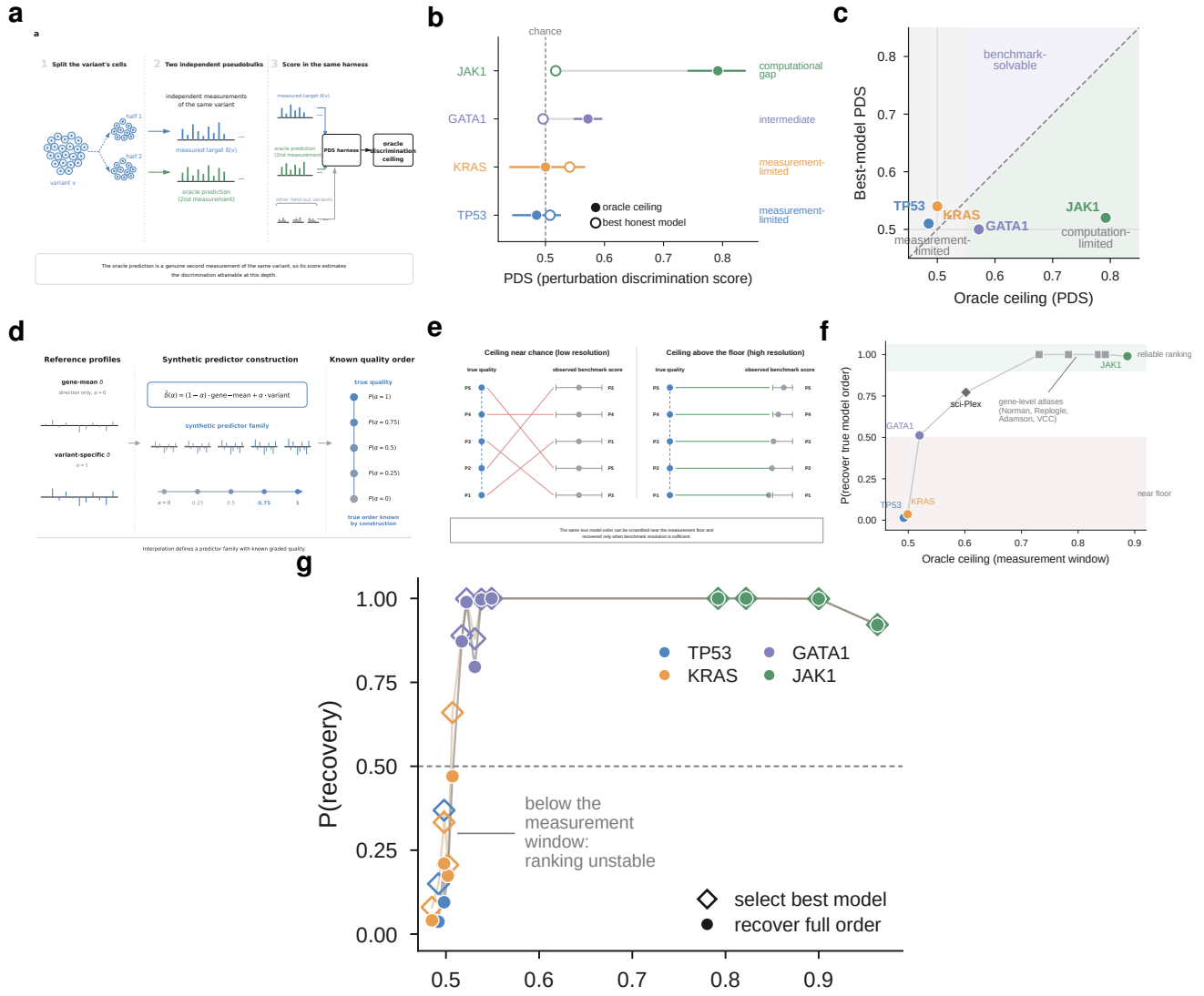


Figure 5: Replicate resolution governs attainable discrimination and benchmark validity. **a**, The oracle predictor defined: an independent second split-half measurement of each held-out variant, scored through the identical harness, estimating the discrimination attainable at the available depth. **b**, Oracle ceiling per gene (native depth) versus the best in-house model. TP53 and KRAS ceilings are at chance (measurement-limited); the best model's slight excess over the ceiling reflects best-of-18-heads selection noise, its interval also overlapping 0.50, not a real gap. The JAK1 ceiling is high while models remain near chance (a residual computational gap). **c**, Ceiling-versus-model phase map separates two failure modes: TP53 and KRAS are measurement-limited (both near chance), JAK1 is computation-limited (ceiling high, model low), and the upper-right benchmark-solvable regime is where a model would track its ceiling. **d**, Construction of synthetic predictors of known, graded quality by interpolating between the gene-mean profile and each variant's measured effect, fixing the true ordering by design. **e**, Illustration of benchmark validity: the same true model order is scrambled by a near-floor benchmark but recovered by a higher-resolution one. **f**, Benchmark resolution across nine perturbation datasets: the probability of recovering the true ranking of graded synthetic predictors (resolution) versus each dataset's oracle ceiling (estimated at a matched 50 cells per half). The four allele genes and five external gene- and drug-level screens trace a phase transition, collapsing at the floor (TP53, KRAS) and approaching one above a ceiling of ~ 0.65 . **g**, The benchmark-validity transition: full-order recovery and best-model selection probabilities are both low below the resolution threshold (TP53, KRAS) and jump toward one once the ceiling clears ~ 0.57 (GATA1) and beyond (JAK1); selecting a winner leads recovering the full order.

creased to 39% and Norman to 11%. This pattern indicates that depth normalization can substantially change which perturbations are evaluable, while dataset-specific effect-size structure remains important. Together, these findings motivate a power-aware workflow for variant-level perturbation evaluation (Fig. 6g): estimate effect size and the split-half floor in a pilot assay; triage variants by predicted rankability; size the experiment to clear the effect-size-to-noise floor; and benchmark models only on perturbations whose ground truth is sufficiently measurable. This converts variant-level perturbation evaluation from uniform model ranking into a power-aware, stratified evaluation design.

Discussion

AllelePerturb shows that allele-specific perturbation prediction is not simply gene-level response prediction at finer resolution. At the allele level, recovery of a shared perturbation program and identification of a specific variant become separable objectives. Models can reproduce the overall transcriptional direction associated with perturbing a gene while carrying little information about which coding variant generated that response. A high correlation with the measured response therefore does not, by itself, establish allele-specific prediction; it may instead reflect a gene-shared program that is recoverable even without informative variant features. Allele-level prediction must consequently be defined jointly by the computational mapping from variant features to cellular response and by the resolution of the experimental measurement used to evaluate that mapping.

This leads to a three-level evaluation hierarchy: detection, identification and prediction. Detection asks whether a perturbation differs from its reference condition, identification asks whether related perturbations can be reproducibly distinguished from one another, and prediction asks whether the response to an unseen perturbation can be inferred from its molecular features. These questions are ordered: detection does not guarantee identification, and prediction cannot be meaningfully evaluated when identification is unsupported by the measurement. This distinction separates measurement-limited from computationally unresolved regimes. For TP53 and KRAS, above-null allele discrimination could not be reliably demonstrated under the available measurement and scoring protocol, whereas JAK1 contained reproducible allele-specific structure that the evaluated models did not recover. The latter therefore defines an evaluable but unresolved regime rather than evidence that allele-level prediction is intrinsically impossible.

These findings have broader implications for virtual-cell benchmarking. Existing benchmarks commonly treat experimentally derived response profiles as fixed ground truth and place uncertainty primarily on the model^{1,16–18}. Recent work has shown that rankings can also depend on baseline choice, systematic variation and metric geometry^{25,26}; AllelePerturb identifies measurement resolution as an additional and logically prior source of uncertainty. A benchmark can compare models reliably only when its measurements preserve the distinctions that define the task. Below that resolution, increasing model complexity cannot create evaluable information, and leaderboards may be unstable even when predictors differ in underlying quality. Benchmark design should therefore distinguish perturbation-level resolution, which determines whether individual interventions are reproducibly separable, from benchmark-level ranking resolution, which determines whether the perturbation set as a whole can recover differences among predictors. This principle extends beyond coding variants to related drugs, doses, combinations, weak genetic perturbations and subtle cell-state transitions.

The results also imply that experimental design and computational evaluation should be treated as a coupled problem. Resolution depends not only on nominal cell depth, but on effect magnitude, inter-allele separation, biological heterogeneity, technical noise, variant assignment and cellular context. A small pilot can identify perturbations that are likely to benefit from deeper profiling and those for which additional depth alone may be insufficient. In the latter setting, more informative experimental designs may be required, including biological replication, improved variant assignment, matched stimulation, responsive cell states, time-resolved sampling or orthogonal measurements of chromatin, protein abundance, signaling activity and phenotype. Similarly, more expressive protein representations cannot compensate for an unresolved transcriptional target, while a resolvable target does not guarantee that current

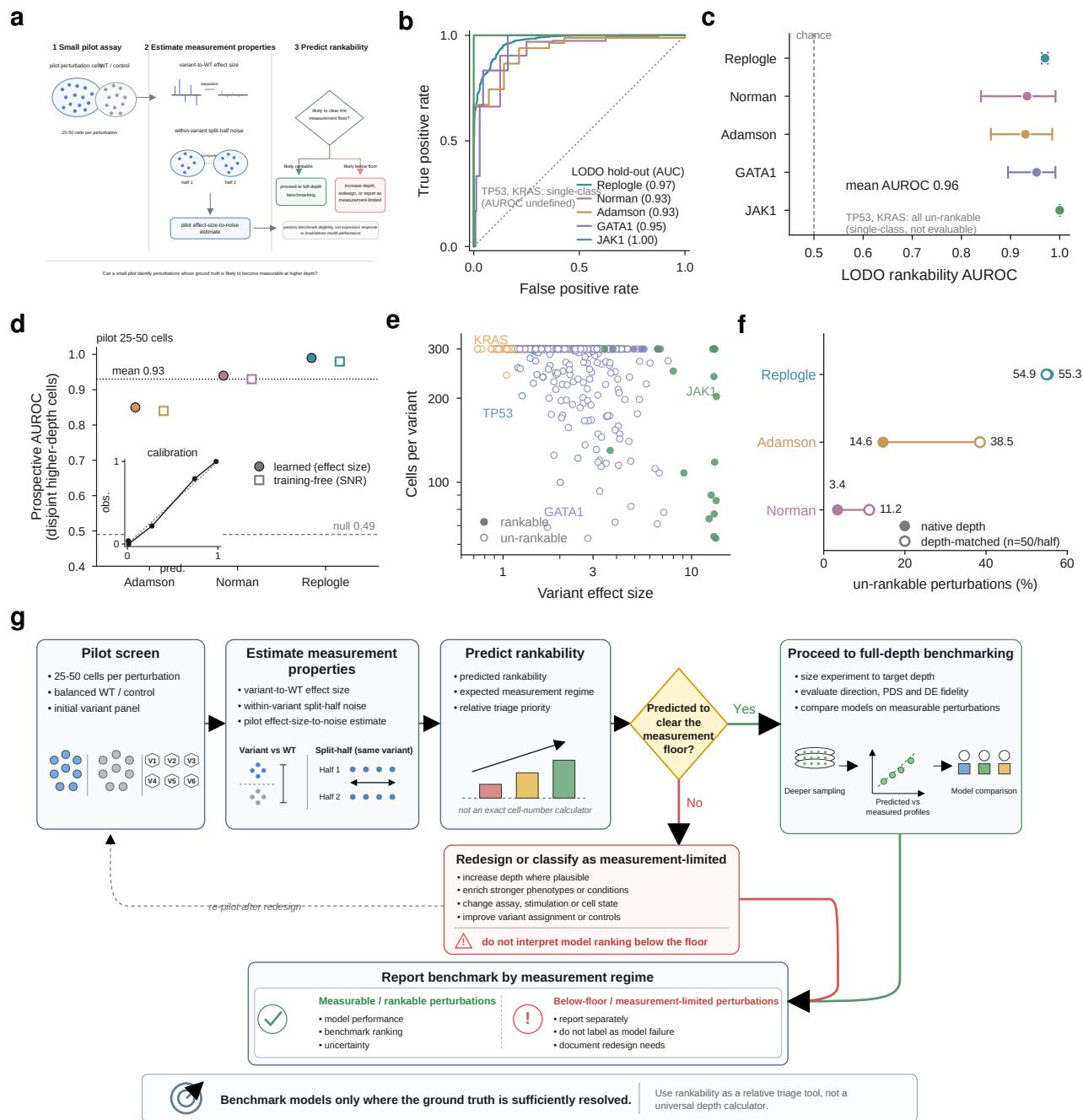


Figure 6: A power-aware reporting protocol for allele-resolution perturbation prediction. **a**, Rankability-prediction pipeline: a pilot-estimable effect size triages which perturbations are evaluable before full model comparison. **b**, Leave-one-dataset-out ROC curves of the single-feature (log effect-size) rankability predictor. **c**, Dataset-level LODO AUROC (mean 0.96): Replogle 0.97, Norman 0.93, Adamson 0.93, GATA1 0.95, JAK1 1.00; TP53 and KRAS are uniformly un-rankable at native depth (single class), so their AUROC is undefined and they are not evaluable. **d**, Prospective validation: from a 25 to 50-cell pilot, rankability at higher depth is predicted on disjoint held-out cells under leave-one-dataset-out cross-validation (AUROC 0.85, 0.94 and 0.99 for Adamson, Norman and Replogle; label-permutation null 0.49), with near-diagonal calibration. **e**, Effect-size-conditioned design landscape: rankability is set by the effect-size regime, not by a universal cell-count cutoff (filled, rankable; open, un-rankable). **f**, Native versus depth-matched ($n = 50$ per half) un-rankable fraction on external gene-level atlases: Replogle stays high (55%) while Adamson (15% to 39%) and Norman (3% to 11%) rise toward it, indicating dataset-specific effect-size structure beyond depth alone. **g**, Recommended power-aware allele-resolution workflow.

sequence- or feature-based models capture the context-dependent mapping from molecular variation to cellular response. Future allele-conditioned models will therefore need to connect protein structure and biochemical mechanism with pathway state and cellular context, but they must be evaluated in datasets with independently reproducible allele-specific responses.

Several limitations define the scope of the present study. The available allele-resolved datasets cover four genes, with gene identity partly confounded by assay, cell type and experimental condition; the observed differences should therefore be interpreted as dataset-specific measurement regimes rather than intrinsic rankings of allele biology. The current prospective triage analysis primarily estimates perturbation-versus-reference resolution, which is necessary but not sufficient for direct power analysis of within-gene allele identification. The replicate-based oracle is an empirical reproducibility reference for a specified depth and scoring protocol, not a mathematical upper bound on all denoising or hierarchical models, and pseudobulk evaluation may underrepresent effects confined to rare subpopulations or changes in cell-state composition. Controlled studies profiling multiple genes within shared experimental contexts, with biological replicates, depth titrations and multimodal readouts, will be needed to establish identification-aware power criteria. More broadly, AllelePerturb establishes a measurement-aware principle for predictive cellular modeling: model performance is interpretable only at biological resolutions that the experiment itself can reproduce. As virtual-cell models move toward variants, combinations and context-specific interventions, progress will depend not only on more expressive architectures, but also on benchmarks designed to distinguish the effects those architectures are asked to predict.

Methods

Study design

AllelePerturb is a retrospective benchmark of held-out protein-coding variant prediction built from publicly available single-cell perturbation datasets; no new human, animal or biological samples were generated. The primary unit of prediction and of statistical resampling is the protein-coding variant condition; throughout, *allele* denotes this experimentally assigned protein-coding variant condition rather than a phased genomic haplotype. Four genes (TP53, KRAS, GATA1 and JAK1) assayed by three technologies (Perturb-seq, base editing and scSNV-seq) were included because they are, to our knowledge, the publicly available single-cell perturbation datasets that resolve individual protein-coding variants with reusable per-variant labels; all eligible variants and cells were used, so sample size was not predetermined. For each generalization split, models were trained on the training variants of a gene and evaluated on its held-out variants, and every model was scored through a single evaluation harness. The primary analyses quantify transcriptional direction recovery and within-gene allele discrimination; separate replicate-based analyses assess perturbation-versus-wild-type detection, within-gene allele identification and benchmark-level model-ranking resolution. Sensitivity analyses varied the expression representation, distance metric, sampling depth and preprocessing protocol.

Benchmark construction

Data sources. AllelePerturb-Bench integrates single-cell perturbation data for four protein-coding genes assayed across three experimental technologies. TP53 (98 variants) and KRAS (92 variants) were obtained from a Perturb-seq screen of coding variants in A549 lung adenocarcinoma cells (Gene Expression Omnibus GSE161824)²³. GATA1 (254 variants) was obtained from a base-editing screen in human hematopoietic stem and progenitor cells, using the processed expression matrix and predefined hold-out splits distributed with PerturbNet²⁴ (Hugging Face repository cyclopeta/PerturbNet_reproduce). JAK1 (26 variants) was obtained from a single-cell single-nucleotide-variant sequencing (scSNV-seq) experiment in HT-29 colorectal cancer cells stimulated with interferon-gamma (European Nucleotide Archive PRJEB48915; processed objects from Zenodo record 10418435)²⁷. Cell counts include both variant-assigned cells and the dataset-specific wild-type or control cells retained after preprocessing; the four datasets together comprise 470 protein-coding variants and 321,043 cells (the latter including

wild-type and control cells).

Preprocessing. For TP53 and KRAS, processed count matrices, cell barcodes and variant-to-cell assignment tables were loaded from GEO, and cells were assigned to variants using the published assignment files, giving a shared 939-gene expression space for the two A549 datasets. For GATA1, we used the PerturbNet-distributed AnnData object, which provides log-normalized expression over 2,477 highly variable genes together with five predefined holdout partitions. For JAK1, the SingleCellExperiment object was loaded in R (Bioconductor); cells were filtered to homozygous genotyped calls with unambiguous coding consequences (missense, splice-site, stop-gained or synonymous), the 2,000 most highly variable genes were selected with `scrani::modelGeneVar` followed by `getTopHVGs`, and log-normalized counts were exported. Each dataset was processed in its own gene space, because the three cell types do not share a common highly-variable-gene set; the object shared across datasets is the variant feature representation, not the gene space. Within each dataset, every gene was standardized to zero mean and unit variance across all cells, and wild-type cells served as the common reference. This standardization is transductive; refitting it on training variants and wild-type cells only, excluding held-out variants, left every predictor at or below chance ($\text{PDS-cosine} \leq 0.48$, no predictor above 0.50) with essentially unchanged direction recovery, and if anything lowered model scores (mean $\Delta\text{PDS} = -0.01$), so the reported all-cell standardization is conservative with respect to the null (Supplementary Note).

Variant feature vector. Each variant v was represented by protein-level features rather than by a variant identifier, so that held-out variants unseen in training can still be predicted. The primary representation is a six-dimensional biophysical vector

$$\theta_v = (\Delta H_v, \Delta V_v, \Delta Q_v, f_v, s_v, h_v),$$

where the first three components are the mutant-minus-wild-type differences in Kyte-Doolittle hydrophobicity, residue molecular weight and formal charge at pH 7.0. Writing $p(\cdot)$ for a residue property and $a_v^{\text{mut}}, a_v^{\text{wt}}$ for the mutant and wild-type amino acids of variant v , each biophysical difference is $\Delta p_v = p(a_v^{\text{mut}}) - p(a_v^{\text{wt}})$, standardized across all 470 variants by $z(\Delta p_v) = (\Delta p_v - \mu_{\Delta p}) / \sigma_{\Delta p}$. The remaining three components are structural indicators. The fold-core term f_v was set to 1.0 for residues within an annotated structural core (the DNA-binding, G-, zinc-finger and kinase domains of TP53, KRAS, GATA1 and JAK1, respectively), to 0.5 for annotated interaction regions, and to 0.0 elsewhere. The functional-switch term s_v was set to 1.0 for catalytic or metal-coordinating residues. The hotspot or pathogenic marker h_v was defined exclusively from external prior knowledge (canonical COSMIC and IARC missense hotspot codons, ClinVar pathogenic annotations and gene-specific functional-module membership), with no reference to the single-cell outcome; the full per-gene residue and codon assignments are listed in Supplementary Table 1. An earlier version of h_v used the top quintile of measured effect size, which leaks the evaluation outcome into the feature; we removed it and re-ran the entire evaluation grid. The ranking result was unchanged after this correction (θ -only PDS shifted by at most 0.02 and every method remained at chance; Supplementary Information).

Protein-language-model embeddings. As an alternative representation, we extracted per-variant embeddings from ESM-1v (`esm1v_t33_650M_UR90S_1`)³³. Writing $\mathbf{H}_v \in \mathbb{R}^{L_v \times 1280}$ for the final-layer hidden states of the full-length mutant sequence of length L_v , the embedding is the mean over residue positions,

$$e_v = \frac{1}{L_v} \sum_{\ell=1}^{L_v} \mathbf{H}_v[\ell, :] \in \mathbb{R}^{1280},$$

used directly, without dimensionality reduction, as the ESM feature space.

Evaluation protocol (AllelePerturb-Eval)

Pseudobulk perturbation profiles. Let C_v and C_{WT} denote the cells of variant v and of wild type, and $x_i \in \mathbb{R}^G$ the standardized expression of cell i over G genes. The perturbation profile of variant v is the pseudobulk difference

$$\delta_v = \frac{1}{|C_v|} \sum_{i \in C_v} x_i - \frac{1}{|C_{WT}|} \sum_{j \in C_{WT}} x_j.$$

To bound the influence of the most deeply sampled variants, up to 300 cells were sampled without replacement per variant and per wild-type reference; variants with fewer than five cells were excluded. Every predictor maps a variant feature vector to an estimate $\hat{\delta}_v$ of this profile, and all metrics compare $\hat{\delta}_v$ to the measured δ_v . The prediction target throughout is this population-average profile; the split-half distributional analyses introduced below are diagnostics of measurement resolution on the single-cell distributions, not alternative prediction targets.

AllelePerturb-Eval separates prediction quality into three families of metrics. **Direction recovery** asks whether a predicted profile points the right way in expression space. **Allele discrimination** asks whether a predicted profile is specific enough to identify its own variant among the other held-out variants. **Differential-expression fidelity** asks whether the predicted and measured profiles agree on which genes change and by how much. The ten metrics are defined below; unless noted, each is computed per held-out variant and then averaged over the held-out set T (of size $n_{\text{test}} = |T|$) of a gene, over genes and over splits.

Perturbation discrimination score (PDS). PDS is the primary metric and directly tests allele discrimination: is a predicted profile closer to its own measured profile than to those of the other held-out variants? This nearest-match discrimination logic follows the perturbation-discrimination principle used in recent virtual-cell evaluation efforts^{1,26}, adapted here to protein-coding variants and computed with tie-aware ranking. For a held-out variant v of gene g , the predicted profile $\hat{\delta}_v$ is ranked by distance $d(\cdot, \cdot)$ against the measured profiles of all candidate variants of the same gene, $C_g = \text{train}(g) \cup \text{test}(g)$ (both training and held-out variants passing the five-cell filter), with $n = |C_g|$. Writing $\ell_v = |\{w \in C_g : d(\hat{\delta}_v, \delta_w) < d(\hat{\delta}_v, \delta_v)\}|$ for the number of candidates strictly closer than the true match and $e_v = |\{w \in C_g : d(\hat{\delta}_v, \delta_w) = d(\hat{\delta}_v, \delta_v)\}| \geq 1$ for the number tied with it (including v itself), the tie-aware score is

$$\text{PDS}(v) = 1 - \frac{\ell_v + \frac{1}{2}(e_v - 1)}{n - 1},$$

so the nearest match scores 1 and random matching 0.5 in expectation, with ties assigned the average (mid) rank. Only held-out variants are scored, but each competes against the full same-gene candidate set, so a correct prediction must be closer to its own variant than to every other measured variant of the gene, held out or training. A prediction with $\|\hat{\delta}_v\| = 0$, or a gene with a single candidate ($n \leq 1$), is assigned 0.5 by definition.

PDS was computed under three distances,

$$d_{\cos}(a, b) = 1 - \frac{a^\top b}{\|a\| \|b\|}, \quad d_1(a, b) = \|a - b\|_1, \quad d_2(a, b) = \|a - b\|_2,$$

giving PDS-cosine (primary), PDS-L1 and PDS-L2. The WT-null reference predicts $\hat{\delta}_v = 0$ for every variant and therefore scores exactly 0.5 by the zero-prediction rule above (its cosine distance to a zero vector is undefined and is not evaluated). Because the empirical chance level under small, gene-stratified candidate sets with ties can depart slightly from the analytic 0.5, we additionally estimated it by label permutation (see Statistical analysis).

Direction-recovery metrics. Averaging the Pearson correlation over held-out variants gives

$$\text{Pearson-}\delta = \frac{1}{n_{\text{test}}} \sum_{v \in T} \text{corr}(\hat{\delta}_v, \delta_v),$$

and δ -cosine is the analogous average of $\hat{\delta}_v^\top \delta_v / (\|\hat{\delta}_v\| \|\delta_v\|)$. Pearson- δ (top-20) restricts the correlation to the 20 genes with the highest variance across the training-set profiles, testing whether agreement is concentrated in a small number of high-variance genes.

Differential-expression metrics. For each variant, differentially expressed genes were defined by a per-gene Welch t -test of variant against wild-type cells (up to 300 cells per group). Let $\mathcal{D}_v^{\text{true}}$ be the $k = 50$ genes with the largest absolute t -statistic and $\mathcal{D}_v^{\text{pred}}$ the k genes with the largest absolute predicted change. DE overlap is

$$\text{DE overlap}(v) = \frac{|\mathcal{D}_v^{\text{pred}} \cap \mathcal{D}_v^{\text{true}}|}{k}.$$

DE-LFC Spearman is the Spearman correlation between the measured log-fold-changes $\ell_{v,g}$ and the predicted changes $\hat{\delta}_{v,g}$ over $\mathcal{D}_v^{\text{true}}$, and direction agreement is the fraction of genes in $\mathcal{D}_v^{\text{true}}$ for which $\text{sign}(\hat{\delta}_{v,g}) = \text{sign}(\ell_{v,g})$. Both quantities use the Welch-test log-fold-change $\ell_{v,g}$ from the differential-expression subsample as the measured reference, computed independently of the pseudobulk profile δ_v .

Reconstruction metric. Mean absolute error is $\text{MAE}(v) = G^{-1} \sum_g |\hat{\delta}_{v,g} - \delta_{v,g}|$, averaged over held-out variants.

Generalization splits

Six split strategies probe distinct axes of generalization. In each, models are trained on the training variants of a gene and evaluated on its held-out variants.

1. Random: the predefined per-dataset holdout partition shipped with AllelePerturb-Bench (GATA1 uses the PerturbNet partition); the realized held-out sets are 16/98 (TP53), 8/92 (KRAS), 52/254 (GATA1) and 9/26 (JAK1) variants. This approximates a random holdout, but the per-gene fraction varies with the source partition rather than being fixed at a single value.
2. Positional (out-of-distribution position): variants in the C-terminal half of each protein held out.
3. Mechanistic (out-of-distribution mechanism): variants at hotspot or functional-switch residues held out.
4. Cross-gene: all GATA1 and JAK1 variants held out, with training on the TP53 and KRAS variants. This split is defined for completeness but excluded from the main grid, because the datasets do not share a gene space and cross-gene transfer is therefore not defined for a shared decoder (see Discussion).
5. Low-depth: variants with fewer than 200 cells held out, isolating the shallow-sampling regime; only GATA1 (45 variants) and JAK1 (17) contain such variants, because all TP53 and KRAS variants exceed this depth.
6. Compatibility: the GATA1 held-out set matched to the PerturbNet evaluation protocol (the same 52 variants as the random split), together with the predefined TP53 (10) and KRAS (8) holdouts, enabling comparison with prior work.

Prediction methods

We evaluated 20 predictors: 18 feature-model combinations formed by crossing six regression heads with three feature spaces, plus two references. Each predictor learns a map $g_\phi : \mathbb{R}^p \rightarrow \mathbb{R}^G$ from a p -dimensional variant feature vector to a G -dimensional pseudobulk profile, fit by minimizing squared error on the training variants, $\min_\phi \sum_{v \in \text{train}} \|g_\phi(\text{feat}_v) - \delta_v\|_2^2$ (with the head-specific regularizer below), and scored by the identical protocol above.

Regression heads. We used six standard regression heads spanning linear, tree-ensemble, neighbor and neural-network families: ridge, lasso, random forest, gradient-boosted trees, k -nearest-neighbors regression and a multilayer perceptron (hyperparameters in Supplementary Table 2). Linear and neighbor heads regress directly onto the full expression profile. Because tree ensembles scale poorly to thousands of output genes, for output spaces larger than 200 genes the random forest and gradient-boosted heads

were fit on a q -component PCA of the training profiles, with predictions mapped back to gene space by the inverse transform. All heads used a fixed random seed.

Feature spaces. θ (six-dimensional biophysical vector); ESM (1,280-dimensional ESM-1v embedding); and ESM+ θ (their 1,286-dimensional concatenation). As a representation robustness control (Supplementary Note) we additionally extracted ESM2-650M³⁰ final-layer representations of the mutant sequence, three mutation-aware constructions of the mutant-minus-wild-type difference (the global mean difference, the mutated-residue hidden-state difference $h_p^{\text{mut}} - h_p^{\text{wt}}$, and a ± 16 -residue local-window difference; proteins longer than 1,022 residues were cropped to a window centered on the mutation, and multi-substitution variants averaged their site differences), and AlphaMissense³¹ pathogenicity scores; all were scored through the identical splits, heads and metrics.

References. The WT-null reference predicts $\hat{\delta}_v = 0$ for every variant, quantifying the score attainable with no perturbation signal. The Gene-mean reference predicts the mean training-variant profile of the held-out variant’s own gene, $\hat{\delta}_v = |\text{train}(g_v)|^{-1} \sum_{w \in \text{train}(g_v)} \delta_w$; because it uses the gene label g_v , it has access to information a variant-blind predictor does not, and we report it as a gene-identity reference on how much of the response is explained by gene identity alone rather than as a competing method. Gene-mean is defined only for genes with at least one training variant in a split and is therefore omitted under the cross-gene split.

Published models under a unified evaluation. To compare against published perturbation-prediction methods, every model was scored through a single evaluation harness: the same expression arrays, deterministic pseudobulk profiles, tie-aware PDS and bootstrap and permutation procedures used for the in-house heads, across the five main splits (Supplementary Table 4). Each model was trained per gene on the training variants and their wild-type control and predicted each held-out variant from its feature vector; held-out variant response cells were never used in training. Because the published scGen and scVIDR packages condition on a categorical perturbation label rather than a continuous per-variant feature, we used in-house re-implementations of their latent vector arithmetic following refs^{4,34}: a per-(gene, split) variational autoencoder with θ as the perturbation attribute and the predicted latent shift decoded onto wild-type cells, scVIDR additionally scaling the shift by a per-variant severity term. Biolord³⁵ (official biolord 0.0.3, ordered-attribute disentanglement) and CellFlow³⁶ (official cellflow-tools, conditional optimal-transport flow) were run from their released packages, each variant’s ESM-1v embedding serving as the CellFlow condition, including held-out variants so unseen alleles resolve by feature. We additionally adapted CPA⁵ (official cpa-tools 0.8.8) in a chemCPA style, replacing its categorical perturbation dictionary with a frozen encoding of the variant features; this model produced non-finite training losses on the continuous inputs and did not converge, even with gradient clipping and reduced learning rates, and is reported for interface completeness rather than scored. Three further models (scGPT⁸, GEARS⁷ and STATE³⁷) condition on perturbed-gene identity and assign an identical prediction to every variant of a gene; we therefore report their allele coverage (4 of 470 variants) rather than a discrimination score, which is undefined for a gene-invariant predictor. PerturbNet²⁴ (conditioned on ESM-1v) generates within a 50-dimensional principal-component subspace fit on wild-type cells; because random predictions confined to this subspace already score above the analytic chance level, we evaluated it against a permutation null estimated in the same subspace, under which its exceedance was not significant (subspace-inflation controls and values in Supplementary Table 4 and results/canonical/perturbnet_subspace_test.json). Confidence intervals are 2,000-sample bootstraps over held-out variants; permutation nulls use 1,000 prediction-to-target shuffles, as for the in-house grid.

Split-half detection analysis

To separate model limitation from measurement limitation, we asked whether variant responses are stably measurable at the available sampling depth. The energy distance between two cell sets $X = \{x_i\}_{i=1}^n$ and $Y = \{y_j\}_{j=1}^m$ is

$$E(X, Y) = \frac{2}{nm} \sum_{i=1}^n \sum_{j=1}^m \|x_i - y_j\| - \frac{1}{n^2} \sum_{i,i'} \|x_i - x_{i'}\| - \frac{1}{m^2} \sum_{j,j'} \|y_j - y_{j'}\|,$$

with Euclidean norms. For each variant with at least 50 cells, cells were partitioned into two random halves A and B , and a size-matched wild-type control sample C was drawn. We computed the within-variant replicate distance $D_{\text{self}} = E(A, B)$ and the variant-to-wild-type distance $D_{\text{null}} = E(A, C)$. As a control on the reference, we additionally computed the wild-type-to-wild-type distance D_{wtwt} from an independent split of the wild-type cells; for TP53, KRAS and GATA1 this equaled D_{null} ($D_{\text{wtwt}}/D_{\text{null}} = 0.91\text{--}1.09$), so the detection floor reflects the absence of variant signal rather than control heterogeneity, whereas for JAK1 D_{null} exceeded D_{wtwt} eight- to thirty-fold (Supplementary Note). Distances were evaluated over a range of subsample depths per half and in three representation spaces, with 50 random seeds per configuration; the canonical values use a 50-component PCA at native maximum depth (Supplementary Table 2). Each seed distribution was summarized by its median and its 2.5 to 97.5 percentile interval.

A variant was classified as detection-eligible (labeled *rankable* in the figures) when its variant-to-wild-type signal exceeded the sampling width of its replicate distance,

$$S = \text{med}(D_{\text{null}}) - \text{med}(D_{\text{self}}) > W = Q_{97.5}(D_{\text{self}}) - Q_{2.5}(D_{\text{self}}),$$

and un-rankable otherwise. This signal-versus-noise criterion is the primary definition; we confirmed that the conclusions are qualitatively unchanged under four alternatives (lower confidence bound of S above zero; $S > 1.5W$; $S > 2W$; and a ratio-based threshold), which shift absolute fractions but preserve the ordering across datasets (Supplementary Fig. 1). The same analysis was applied to the four AllelePerturb genes and to three widely used gene-level Perturb-seq atlases, Replogle 2022², Norman 2019²⁹ and Adamson 2016²⁸ (K562 cells), obtained from the scPerturb data portal¹⁶, using an identical native-depth, all-perturbation, direct split-half definition throughout. Here native depth denotes each perturbation evaluated at the deepest split-half subsample rung it supports (from 25 to 800 cells per half) in the 50-component PCA space, so that shallower datasets are scored at correspondingly shallower depth; the resulting per-dataset rankability results are those reported in Fig. 3h (rankable fraction) and Fig. 6f (un-rankable fraction).

Rankability predictor

To test whether detection-eligibility can be anticipated before a full model comparison, we trained a logistic-regression classifier to predict per-perturbation rankability (the detection-level split-half label above) under leave-one-dataset-out cross-validation. Each perturbation contributed a single example: its canonical native-depth rankability label (deepest available split-half bin, energy distance, PCA space, as above) and its pilot-estimable pseudobulk effect size $\|\delta_v\|_2$. We excluded the split-half quantities that define the label (S , W , their ratio, D_{self} , D_{null}) and any evaluation-configuration identity (distance metric, representation space, subsample depth), which are not properties of the perturbation and would leak the label. Discrimination was measured by the area under the receiver-operating-characteristic curve with 2,000-sample bootstrap intervals over held-out perturbations; held-out datasets whose label was single-class (all un-rankable, namely TP53 and KRAS at native depth) are reported as not evaluable rather than assigned an undefined AUROC. The predictor is implemented in `scripts/figures/rankability_predictor.py`.

Prospective pilot-to-full validation. To test that the rankability predictor is prospective rather than a retrospective consequence of same-source data, we re-evaluated it on cells disjoint from those defining the label. For each perturbation a single random permutation assigned the first 50 cells to a pilot set and the next $2T$ cells to an evaluation set; the pilot effect size $\|\hat{\delta}\|_2$, and a pilot signal-to-noise ratio from a pilot split-half, were computed from 25 or 50 pilot cells, while the rankability label at target depth $T \in \{100, 200\}$ was computed by the split-half criterion on the disjoint evaluation cells. A leave-one-dataset-out logistic predictor and a training-free pilot signal-to-noise threshold were evaluated across

the seven datasets; the rankability label is computed only on evaluation cells disjoint from the cells used to form the pilot features, so this is a prospective test. We report per-dataset AUROC over the datasets with balanced labels rather than a pooled value, because pooling mixes differing per-dataset base rates, and a within-dataset label-permutation control returned a null AUROC of 0.49. Scripts are provided under `results/pilot_validation/`.

Measurement ceiling, pairwise identification and benchmark resolution

Three analyses separate measurement limits from model limits; all reuse the identical deterministic pseudobulk and tie-aware PDS of the evaluation protocol, average over 15 random cell subsamples, and require no additional data.

Oracle ceiling. For each variant with at least five cells, cells were partitioned into two disjoint halves and a size-matched wild-type control was split likewise, giving two independent pseudobulk deltas. Using one half as the prediction and the other as the evaluation truth, we scored PDS through the identical harness (same candidate sets, tie-aware ranking and split structure as the model comparison). Because the prediction is an independent second measurement of the same variant, its PDS defines an empirical replicate-based discrimination ceiling at that depth and protocol, reported as an empirical, depth- and protocol-specific quantity rather than a mathematical upper bound on models that borrow information across variants. PerturbNet is excluded from this per-gene comparison because it is scored in a reduced principal-component subspace that is not comparable to the full-gene-space oracle.

Ceiling sensitivity and gene-shared decomposition. We assessed the sensitivity of the ceiling to the split-half protocol. Recomputing the oracle using all available cells per half rather than the 300-cell cap left it unchanged, and substituting a near-true prediction (the deep mean of a variant from cells disjoint from the evaluation truth) scored against the benchmark’s noisy pseudobulk truth remained at chance for the low-effect genes; shrinking predictions toward the gene mean did not raise discrimination. These sensitivity outcomes are reported with the corresponding results (Supplementary Note). Separately, we decomposed each effect into a gene-shared mean and an allele-specific residual, $\delta_v = \bar{\delta}_g + r_v$, and recomputed the oracle and the per-predictor Pearson- δ on the residual profiles to quantify how much of direction recovery is gene-shared.

Pairwise identification. In the same cosine-delta space, we computed the split-half replicate distance $D_{\text{self}}(v)$ and the pairwise distance $D(v_i, v_j)$ between variant mean deltas at a matched half-depth. A pair was called resolvable when $D(v_i, v_j)$ exceeded the mean of the two variants’ replicate distances, and a variant identifiable when its nearest sibling lay beyond its own replicate noise.

Metric-family and classifier controls. To test whether the result depends on the energy-distance metric, we recomputed the split-half window ratio $D_{\text{self}}/D_{\text{null}}$ in the 50-component PCA space under energy distance, maximum-mean-discrepancy (RBF kernel, median-heuristic bandwidth), sliced-Wasserstein, cosine and Euclidean distances (Supplementary Note). As a distribution-free identification test, we trained an ℓ_2 -regularized logistic-regression classifier to separate the single cells of two conditions under five-fold cross-validation in the same space, with the principal components fit label-blind and classes balanced by subsampling; we report the area under the ROC curve for variant-versus-variant (identification) and variant-versus-wild-type (detection) comparisons. Because all variants of a gene derive from one pooled experiment, there is no per-variant batch structure; label-permutation and wild-type-versus-wild-type controls both returned a chance AUROC of 0.5, and where the classifier exceeded chance its accuracy survived regressing out per-cell depth.

Benchmark resolution. To test whether the oracle ceiling governs benchmark validity, we built synthetic predictors of known, graded quality, $\hat{\delta}(v, \alpha) = (1 - \alpha) \bar{\delta}_g + \alpha \delta_v^{\text{build}}$, where δ_v^{build} is a pseudobulk delta from a build subsample disjoint from the evaluation truth and $\bar{\delta}_g$ is the gene mean; true quality increases monotonically with α , and at $\alpha = 1$ the attainable score equals the oracle ceiling. Scoring the family through the harness and bootstrapping the held-out variants, we recorded the probability of recovering the full α -ordering and of selecting the top predictor, as a function of the oracle window and of the number of held-out variants.

External benchmark calibration. We applied the identical controlled-predictor resolution analysis to five public perturbation datasets (Norman, Replogle, Adamson, the Virtual Cell Challenge training set and sci-Plex³²), loading each one’s ground-truth single-cell profiles and computing the resolution, the probability of recovering the true predictor ordering under bootstrap over perturbations, at 50 cells per half over perturbations with at least 100 cells (Fig. 5f). Because this inclusion threshold and depth differ from the native-depth main analysis, the allele perturbation counts and rankable fractions there differ from Supplementary Table 5. Analysis scripts are provided under `results/benchmark_resolution/`.

Analytical scaling. Under a finite-sample pseudobulk model, two half-profiles of a variant satisfy $\mathbb{E}\|d_A - d_B\|^2 = 2\eta^2$ with per-profile noise $\eta^2 = \text{tr}(\Sigma)/m$, and two variants satisfy $\mathbb{E}\|d_i - d_j\|^2 = \|\mu_i - \mu_j\|^2 + 2\eta^2$, giving $(D_{\text{self}}/D_{\text{null}})^2 = 1/(1 + S)$ with $S = \|\delta\|^2/(2\eta^2)$ and a dimensionless discrimination signal-to-noise $\rho = \bar{\Delta}\sqrt{m}/(\sqrt{2}\eta)$; discrimination depends on the data only through ρ and the number of candidate variants. We confirmed the derived depth scaling empirically ($\eta^2 \propto 1/m$ across all genes). The noise-debiased between-variant signal $\bar{\Delta}^2$ lay at or below the sampling-noise floor for TP53, KRAS and GATA1 at every tested depth, so only JAK1 traced the increasing branch of ρ . The scaling is derived; the functional form is calibrated from the synthetic sweep rather than assumed for the tie-aware cosine score.

Power analysis

To relate detection to sampling depth, cells were subsampled at 50, 100, 150 and 300 per variant and the split-half rankability test was recomputed at each depth. The fraction of variants classified as detectable was plotted against subsample size for each gene, and a detection threshold (the depth at 50% detection) was estimated by sigmoid fitting where the data supported it.

Statistical analysis

Confidence intervals for PDS and Pearson- δ were obtained by nonparametric bootstrap over held-out variants (2,000 resamples, percentile method, 95% level). A method’s PDS was called to differ from chance when its 95% confidence interval excluded 0.50, and its direction significantly positive when the lower bound of its Pearson- δ interval exceeded 0. The per-gene $D_{\text{self}}/D_{\text{null}}$ ratio (Fig. 3c) and the external-atlas rankability fractions (Fig. 6f) were bootstrapped over variants and over perturbations, respectively. To confirm that the analytic PDS chance level of 0.50 holds under the actual tie structure and small test sets, we ran a within-split label-permutation control: for each method-split-gene combination the full prediction-to-truth distance matrix was precomputed and the prediction-to-target correspondence was permuted 1,000 times, recomputing PDS with the same tie-aware ranking each time. The empirical null mean was 0.500 (range 0.495 to 0.504 across 340 combinations), and only 3.2% of combinations exceeded the permutation 95th percentile, consistent with chance under the empirical permutation distribution; because these method-split-gene combinations are correlated, this is a descriptive check rather than a formal false-positive calibration. All bootstraps and permutations used fixed random seeds. Because some residues carry more than one substitution, we also resampled residue-level clusters rather than individual variants; the per-method intervals were essentially unchanged, with every non-null predictor still overlapping chance, and training-to-held-out residue overlap ranged from 0% (positional split) to 54% (compatibility split) without altering the chance-level result (Supplementary Note). We report intervals to characterize uncertainty rather than as per-method hypothesis tests: the conclusions rest on the direction and consistency of the effect across all 19 non-null predictors, which move together, and so do not depend on any single comparison or on a multiplicity correction. These intervals quantify within-benchmark uncertainty over held-out variants, not generalization across the four genes, assays or technologies, which the four-dataset design cannot estimate. To ensure that discrimination estimates did not depend on a single stochastic pseudobulk draw, each variant’s PDS in the method comparison was first averaged over 15 independent random subsamples of cells; the reported point estimate is then the mean over (split, gene) cells of the within-cell mean of these seed-averaged per-variant scores, and the 95% confidence interval is a 2,000-sample nonparametric bootstrap that re-

samples held-out variants within each (split, gene) cell. No statistical method was used to predetermine sample size; all eligible public variants and cells were included, and measurement power was instead assessed retrospectively and prospectively through the split-half and pilot analyses described above.

Software and reproducibility

The evaluation grid was run in Python 3.11.14 with NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2 and SciPy 1.14.1. Single-cell loading and preprocessing used scanpy 1.10.3 (Python) and scanr with SingleCellExperiment (R, Bioconductor). ESM-1v embeddings were extracted with fair-esm 2.0.0 and PyTorch 2.10.0 (CUDA 12.6). Figures were rendered with matplotlib 3.10.8. Published-model adapters were run in their own package environments: Biolord 0.0.3 and the scGen/scVIDR re-implementations under scvi-tools 1.4.3 with PyTorch 2.12; CellFlow (cellflow-tools 0.1.1) under JAX; and CPA (cpa-tools 0.8.8) under scvi-tools 0.20.3 with PyTorch 2.0.

Data availability

All single-cell datasets analyzed in this study are publicly available. The TP53 and KRAS Perturb-seq data are available from the Gene Expression Omnibus under accession GSE161824²³. The GATA1 base-editing data were originally generated by Perturb(BE)-seq in human hematopoietic stem and progenitor cells and deposited at the Gene Expression Omnibus under accession GSE215253; we used the processed expression matrix and predefined holdout splits redistributed with PerturbNet (Hugging Face repository `cyclopet/PerturbNet_reproduce`)²⁴. The JAK1 scSNV-seq data are available from the European Nucleotide Archive under accession PRJEB48915, with processed objects at Zenodo (<https://doi.org/10.5281/zenodo.10418435>)²⁷. The gene-level Perturb-seq atlases used for the rankability analysis (Replogle 2022, Norman 2019 and Adamson 2016) were obtained from the scPerturb data portal¹⁶. The derived AllelePerturb-Bench tables that support all main figures and the Supplementary Information (per-variant pseudobulk profiles, variant feature vectors, generalization-split definitions and the full evaluation grids) are available at <https://github.com/Boom5426/AllelePerturb>.

Code availability

All analysis code, the packaged benchmark (`alleleperturb/`) and the figure and evaluation scripts that regenerate the main-figure and Supplementary values are available at <https://github.com/Boom5426/AllelePerturb>; random seeds are fixed throughout (Statistical analysis). A versioned release will be archived with an assigned DOI on acceptance.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

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